Deal with NaN

In [1]:
```
import pandas as pd
```

In [2]:
```
df=pd.read_csv("wine.csv")
```

In [6]:
```
df2=df
```

In [7]:
```
df2.head()
```

Out[7]:

	Unnamed: 0	country	description	designation	points	price	province	region_1	region_2	
0	0	US	This tremendous 100% varietal wine hails from ...	Martha's Vineyard	96	235.0	California	Napa Valley	Napa	Ca Sau
1	1	Spain	Ripe aromas of fig, blackberry and cassis are ...	Carodorum Selección Especial Reserva	96	110.0	Northern Spain	Toro	NaN	T
2	2	US	Mac Watson honors the memory of a wine once ma...	Special Selected Late Harvest	96	90.0	California	Knights Valley	Sonoma	Sau
3	3	US	This spent 20 months in 30% new French oak, an...	Reserve	96	65.0	Oregon	Willamette Valley	Willamette Valley	Pir
4	4	France	This is the top wine from La Bégude, named aft...	La Brûlade	95	66.0	Provence	Bandol	NaN	Pr re

In [24]:
```
df2[pd.isnull(df2.country)]
```

Out[24]:

	Unnamed: 0	country	description	designation	points	price	province	region_1	region_2
1133	1133	NaN	Delicate white flowers and a spin of lemon pee...	Askitikos	90	17.0	NaN	NaN	NaN
1440	1440	NaN	A blend of 60% Syrah, 30% Cabernet Sauvignon a...	Shah	90	30.0	NaN	NaN	NaN
68226	68226	NaN	From first sniff to last, the nose never makes...	Piedra Feliz	81	15.0	NaN	NaN	NaN
113016	113016	NaN	From first sniff to last, the nose never makes...	Piedra Feliz	81	15.0	NaN	NaN	NaN
135696	135696	NaN	From first sniff to last, the nose never makes...	Piedra Feliz	81	15.0	NaN	NaN	NaN

In [25]:
```
df2.region_2.fillna("Unknown")
```

Out[25]:
```
0                     Napa
1                  Unknown
2                   Sonoma
3         Willamette Valley
4                  Unknown
                 ...
150925             Unknown
150926             Unknown
150927             Unknown
150928             Unknown
150929             Unknown
Name: region_2, Length: 150930, dtype: object
```

In [46]:
```
df2.region_1.replace("Fiano di Avellino", "Avellino")
```

Out[46]:
```
0              Napa Valley
1                     Toro
2           Knights Valley
3        Willamette Valley
4                   Bandol
                ...
150925            Avellino
150926           Champagne
150927            Avellino
150928           Champagne
150929          Alto Adige
Name: region_1, Length: 150930, dtype: object
```

In [47]:
```
df2.dropna()
```

Out[47]:

	Unnamed: 0	country	description	designation	points	price	province	region_1	
0	0	US	This tremendous 100% varietal wine hails from ...	Martha's Vineyard	96	235.0	California	Napa Valley	
2	2	US	Mac Watson honors the memory of a wine once ma...	Special Selected Late Harvest	96	90.0	California	Knights Valley	
3	3	US	This spent 20 months in 30% new French oak, an...	Reserve	96	65.0	Oregon	Willamette Valley	\
8	8	US	This re-named vineyard was formerly bottled as...	Silice	95	65.0	Oregon	Chehalem Mountains	\
9	9	US	The producer sources from two blocks of the vi...	Gap's Crown Vineyard	95	60.0	California	Sonoma Coast	
...	...	...	...	...	...	...	...	...	
150889	150889	US	A bizarre style of wine. The aromas are Port-l...	Lafond Vineyard	82	35.0	California	Santa Ynez Valley	Cer
150892	150892	US	A light, earthy wine, with violet, berry and t...	Coastal	82	10.0	California	California	Califo
150914	150914	US	Old-gold in color, and thick and syrupy. The a...	Late Harvest Cluster Select	94	25.0	California	Anderson Valley	Mendo
150915	150915	US	Decades ago, Beringer's then-winemaker Myron N...	Nightingale	93	30.0	California	North Coast	N

	Unnamed: 0	country	description	designation	points	price	province	region_1
150916	150916	US	An impressive wine that presents a full bouque...	J. Schram	93	65.0	California	Napa Valley

39241 rows × 11 columns

In [54]:
```
df2.dropna(axis=0)
#df2.dropna(axis=1)
```

Out[54]:

	Unnamed: 0	country	description	designation	points	price	province	region_1	
0	0	US	This tremendous 100% varietal wine hails from ...	Martha's Vineyard	96	235.0	California	Napa Valley	
2	2	US	Mac Watson honors the memory of a wine once ma...	Special Selected Late Harvest	96	90.0	California	Knights Valley	
3	3	US	This spent 20 months in 30% new French oak, an...	Reserve	96	65.0	Oregon	Willamette Valley	\
8	8	US	This re-named vineyard was formerly bottled as...	Silice	95	65.0	Oregon	Chehalem Mountains	\
9	9	US	The producer sources from two blocks of the vi...	Gap's Crown Vineyard	95	60.0	California	Sonoma Coast	
...	...	...	...	...	...	...	...	...	
150889	150889	US	A bizarre style of wine. The aromas are Port-l...	Lafond Vineyard	82	35.0	California	Santa Ynez Valley	Cen
150892	150892	US	A light, earthy wine, with violet, berry and t...	Coastal	82	10.0	California	California	Califo
150914	150914	US	Old-gold in color, and thick and syrupy. The a...	Late Harvest Cluster Select	94	25.0	California	Anderson Valley	Mendo
150915	150915	US	Decades ago, Beringer's then-winemaker Myron N...	Nightingale	93	30.0	California	North Coast	N

	Unnamed: 0	country	description	designation	points	price	province	region_1
150916	150916	US	An impressive wine that presents a full bouque...	J. Schram	93	65.0	California	Napa Valley

39241 rows × 11 columns

In [60]:
```
df2.dropna(subset=['region_1'])
```

Out[60]:

	Unnamed: 0	country	description	designation	points	price	province	region_1
0	0	US	This tremendous 100% varietal wine hails from ...	Martha's Vineyard	96	235.0	California	Napa Valley
1	1	Spain	Ripe aromas of fig, blackberry and cassis are ...	Carodorum Selección Especial Reserva	96	110.0	Northern Spain	Toro
2	2	US	Mac Watson honors the memory of a wine once ma...	Special Selected Late Harvest	96	90.0	California	Knights Valley
3	3	US	This spent 20 months in 30% new French oak, an...	Reserve	96	65.0	Oregon	Willamette Valley
4	4	France	This is the top wine from La Bégude, named aft...	La Brûlade	95	66.0	Provence	Bandol
...	...	...	...	...	...	...	...	...
150925	150925	Italy	Many people feel Fiano represents southern Ita...	NaN	91	20.0	Southern Italy	Fiano di Avellino
150926	150926	France	Offers an intriguing nose with ginger, lime an...	Cuvée Prestige	91	27.0	Champagne	Champagne
150927	150927	Italy	This classic example comes from a cru vineyard...	Terre di Dora	91	20.0	Southern Italy	Fiano di Avellino
150928	150928	France	A perfect salmon shade, with scents of peaches...	Grand Brut Rosé	90	52.0	Champagne	Champagne

	Unnamed: 0	country	description	designation	points	price	province	region_1
150929	150929	Italy	More Pinot Grigios should taste like this. A r...	NaN	90	15.0	Northeastern Italy	Alto Adige

125870 rows × 11 columns

GroupBy

In [10]:
```
df2.groupby('province').province.count()
```

Out[10]:
```
province
Achaia                     2
Aconcagua Costa            5
Aconcagua Valley         194
Aegean                    16
Ahr                        5
                        ...
Zenata                     8
Zitsa                      1
Österreichischer Sekt     20
Štajerska                 20
Župa                      11
Name: province, Length: 455, dtype: int64
```

In [13]:
```
df2.groupby('country').points.min()
```

Out[13]:
```
country
Albania                   88
Argentina                 80
Australia                 80
Austria                   81
Bosnia and Herzegovina    83
Brazil                    81
Bulgaria                  80
Canada                    82
Chile                     80
China                     82
Croatia                   81
Cyprus                    80
Czech Republic            85
Egypt                     83
England                   91
France                    80
Georgia                   81
Germany                   80
Greece                    80
Hungary                   80
India                     82
Israel                    80
Italy                     80
Japan                     85
Lebanon                   81
Lithuania                 84
Luxembourg                86
Macedonia                 81
Mexico                    80
Moldova                   81
Montenegro                82
Morocco                   82
New Zealand               80
Portugal                  80
Romania                   80
Serbia                    86
Slovakia                  82
Slovenia                  82
South Africa              80
South Korea               81
Spain                     80
Switzerland               83
Tunisia                   85
Turkey                    84
US                        80
US-France                 88
Ukraine                   83
Uruguay                   80
Name: points, dtype: int64
```

In [15]:
```
df2.groupby(['country']).price.agg([len, min, max])
```

Out[15]:

country	len	min	max
Albania	2.0	20.0	20.0
Argentina	5631.0	4.0	250.0
Australia	4957.0	5.0	850.0
Austria	3057.0	8.0	1100.0
Bosnia and Herzegovina	4.0	12.0	13.0
Brazil	25.0	11.0	35.0
Bulgaria	77.0	7.0	28.0
Canada	196.0	12.0	145.0
Chile	5816.0	5.0	400.0
China	3.0	7.0	27.0
Croatia	89.0	12.0	65.0
Cyprus	31.0	10.0	22.0
Czech Republic	6.0	15.0	25.0
Egypt	3.0	NaN	NaN
England	9.0	38.0	75.0
France	21098.0	5.0	2300.0
Georgia	43.0	9.0	40.0
Germany	2452.0	5.0	775.0
Greece	884.0	7.0	120.0
Hungary	231.0	7.0	764.0
India	8.0	10.0	20.0
Israel	630.0	8.0	150.0
Italy	23478.0	5.0	900.0
Japan	2.0	24.0	24.0
Lebanon	37.0	12.0	51.0
Lithuania	8.0	10.0	10.0
Luxembourg	9.0	36.0	50.0
Macedonia	16.0	12.0	25.0
Mexico	63.0	12.0	108.0
Moldova	71.0	8.0	42.0
Montenegro	2.0	10.0	10.0
Morocco	12.0	6.0	35.0
New Zealand	3320.0	7.0	125.0

country	len	min	max
Portugal	5322.0	4.0	980.0
Romania	139.0	4.0	320.0
Serbia	14.0	15.0	42.0
Slovakia	3.0	15.0	16.0
Slovenia	94.0	7.0	90.0
South Africa	2258.0	5.0	145.0
South Korea	4.0	11.0	16.0
Spain	8268.0	4.0	770.0
Switzerland	4.0	19.0	38.0
Tunisia	2.0	NaN	NaN
Turkey	52.0	14.0	120.0
US	62397.0	4.0	2013.0
US-France	1.0	50.0	50.0
Ukraine	5.0	13.0	13.0
Uruguay	92.0	7.0	60.0

Multilevel indexing

In [16]:
```
countries_reviewed = df2.groupby(['country', 'province']).description.agg([len])
countries_reviewed
```

Out[16]:

country	province	len
Albania	Mirditë	2
Argentina	Mendoza Province	4742
	Other	889
Australia	Australia Other	553
	New South Wales	246
...	...	...
Uruguay	Juanico	19
	Montevideo	3
	Progreso	5
	San Jose	15
	Uruguay	18

455 rows × 1 columns

In [17]:
```
mi = countries_reviewed.index
type(mi)
```

Out[17]: pandas.core.indexes.multi.MultiIndex

In [18]:
```
countries_reviewed.reset_index()
```

Out[18]:

	country	province	len
0	Albania	Mirditë	2
1	Argentina	Mendoza Province	4742
2	Argentina	Other	889
3	Australia	Australia Other	553
4	Australia	New South Wales	246
...	...	...	...
450	Uruguay	Juanico	19
451	Uruguay	Montevideo	3
452	Uruguay	Progreso	5
453	Uruguay	San Jose	15
454	Uruguay	Uruguay	18

455 rows × 3 columns

In [19]:
```
countries_reviewed.sort_values(by='len')
```

Out[19]:

country	province	len
Greece	Central Greece	1
	Zitsa	1
Cyprus	Pafos	1
Slovenia	Slovenska Istra	1
Hungary	Pannon	1
...	...	...
Spain	Northern Spain	4892
France	Bordeaux	6111
Italy	Tuscany	7281
US	Washington	9750
	California	44508

455 rows × 1 columns

In [20]:
```
df2.groupby(['country', 'province']).apply(lambda df: df.loc[df.points.idxmax()]
```

Out[20]:

country	province	Unnamed: 0	country	description	designation	points	price	province	re
Albania	Mirditë	4642	Albania	This garnet-colored wine made from 100% Kallme...	NaN	88	20.0	Mirditë	
Argentina	Mendoza Province	65331	Argentina	If the color doesn't tell the full story, the ...	Nicasia Vineyard	97	120.0	Mendoza Province	M
	Other	10619	Argentina	Take note, this could be the best wine Colomé ...	Reserva	95	90.0	Other	
Australia	Australia Other	68251	Australia	This big wine presents a sophisticated bouquet...	Yattarna	92	65.0	Australia Other	A
	New South Wales	54205	Australia	This wine's deep brassy color suggests honey, ...	Noble One Botrytis	93	32.0	New South Wales	
...	...	...	...	...	...	...	...	...	
Uruguay	Juanico	3160	Uruguay	This mature Bordeaux-style blend is earthy on ...	Preludio Barrel Select Lote N 77	90	45.0	Juanico	
	Montevideo	3164	Uruguay	Bouza ranks as one of Uruguay's top wineries, ...	Monte Vide Eu Tannat-Merlot-Tempranillo	90	57.0	Montevideo	
	Progreso	6541	Uruguay	Blackberry and plum aromas come with wood-smok...	RPF	89	23.0	Progreso	
	San Jose	70157	Uruguay	While this ranks as one of the best Uruguayan ...	El Preciado Premier Gran Reserva	89	60.0	San Jose	
	Uruguay	132482	Uruguay	They call it Special Barrel, and one sniff tel...	Special Barrel	89	50.0	Uruguay	

455 rows × 11 columns